

CV

Basics

Name Luke Lowery
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Website <https://lukelowry.github.io>
Summary Currently employed at Texas A&M University. EE PhD Student of Power Systems

Work

2024 - PRESENT

Research Assistant

Texas A&M University

- Develops Phasor & EMT models to accelerate the integration of renewables in modern transmission systems.
- Performs novel research on the impact of Geomagnetic Disturbances on the nations electric grid.
- Applies modern mathematics to solve open problems in energy and power research.

2023.05 - 2023.08

High Voltage Intern

SpaceX

- Switchgear & transformer installation, design, and coordination for mission-critical loads (5kV & 35kV)
- Responsible for differential and over-current protection on critical feeder lines to production and launch-pad.
- Designed & implemented 500 kW feeder and building distribution to a new production building in 4 weeks.
- Co-designed the microgrid controller functionality with Tesla Engineers. Functionality allowed our Mega-Packs (smart battery storage) to create seamless transitions from the Utility to isolated island operation.

2022.08 - 2023.01

Power Systems Cooperative Education

SpaceX

- Coordinated with other SpaceX production and launch sites to create a standardized SCADA system. Performed transient analysis of synchronization anomalies to discover coordination errors.
- Introduced solar panel expansion to decrease diesel use, turning Starbase into a 50% green energy site.
- Designed a site-wide SCADA system to monitor and automate the site microgrid using Ignition as the front end.

2022.05 - 2022.08

SCADA Intern

SpaceX

- Automated various power distribution tasks such as load-shedding, interactive fuel-consumption projector, and predictive site-load calculator that is still utilized during launch operations.
- Deployed numerous PLCs, control panels, and sensors to increase grid visibility and control using Allen Bradley, Siemens, and SEL hardware. Communication protocols used: MQTT, DNP, Modbus, Profinet, GOOSE.

2021.05 - 2021.08

Control Systems Intern II

ICS Texas

- Designed and implemented control systems for numerous applications including natural gas sites, boiler systems, and production facilities. Primarily completed with Allen Bradley PLC hardware.

2020.05 - 2020.08

Control Systems Intern I

ICS Texas

- Mastered AutoCAD for the design of control panels that were manufactured in-house. Developed an automated AutoCAD program in LISP to synthesize to accelerate manufacturing.

Education

2023 - 2027

 College Station, TX

PhD

Texas A&M University

Electrical Engineering

2019 - 2023

 College Station, TX

Bachelor of Science

Texas A&M University

Electrical Engineering

Leadership

2025.05 - PRESENT

Senator

Graduate and Professional Student Government

Student Senator for the ECEN Graduate Department

- Draft legislation that upholds university values and improves the Aggie experience.
- Vote and represent the Department of Electrical Engineering on University policies and procedures.

Publications

2026

System Frequency Nadir and Trajectory Prediction With Discontinuity Constraints in Governor Dynamics

Jongoh Baek, *Luke Lowery*, and Adam B. Birchfield

IEEE Open Access Journal of Power and Energy, 13, pp. 274–285

DOI: [10.1109/OAJPE.2026.3675094](https://doi.org/10.1109/OAJPE.2026.3675094)

2026	<u>Spectral Graph Filters for Large Power Systems</u> <i>Luke Lowery and Adam B. Birchfield</i> <i>2026 IEEE Texas Power and Energy Conference (TPEC)</i> , pp. 1–6 DOI: 10.1109/TPEC67884.2026.11513110
2026	<u>Using Spectral Graph Wavelets to Analyze Large Power System Oscillation Modes</u> <i>Luke Lowery, Jongoh Baek, and Adam Birchfield</i> <i>59th Hawaii International Conference on System Sciences (HICSS)</i>
2025	<u>EMT Simulation with Spectral Graph Wavelets</u> <i>Luke Lowery and Adam B Birchfield</i> <i>2025 IEEE Texas Power and Energy Conference (TPEC)</i> , pp. 1–6
2025	<u>Voltage Stability in Plausible Non-Uniform Geomagnetic Disturbances</u> <i>Luke Lowery and Adam B. Birchfield</i> <i>IEEE Transactions on Power Systems</i> , pp. 1-12 DOI: 10.1109/TPWRS.2025.3588826
2024	<u>Voltage Stability Interface Limits in Non-Uniform Geomagnetic Disturbances</u> <i>Luke Lowery and Adam B Birchfield</i> <i>2024 56th North American Power Symposium (NAPS)</i> , pp. 1–6

Presentations

 College Station, Texas	<u>Graduate Speaker</u> ECEN Townhall
 College Station, Texas	<u>Graduate Speaker</u> ECEN Fellowship Banquet
 College Station, Texas	<u>Speaker</u> Modern Methods in Electromagnetic Transient Simulations
 College Station, Texas	<u>Speaker</u> Voltage Stability in Non-Uniform Geomagnetic Disturbances

Awards

2025.12.01	<u>Best Paper</u> 59th Hawaii International Conference on System Science <i>Best Paper in Electric Energy Systems Track</i>
2025.11.01	<u>Department Poster Award</u> Texas A&M University <i>2nd Place in Department Poster Event</i>

2024.03.01

Michael Powell Graduate Fellowship
Texas A&M University

2024.03.01

Thomas Powell 1962 Fellowship
Texas A&M University

Projects

2025.12 - 2026.02

Blackout USA
An educational tool for all backgrounds to understand the challenges of operating the electric grid.